

Data Analytics Capstone

Gallstone Disease Risk Prediction Using Machine Learning

Synopsis

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Introduction

Gallstone disease, or cholelithiasis, is a widespread digestive condition in which hardened deposits form in the gallbladder due to imbalances in bile components such as cholesterol or bilirubin. These stones may remain asymptomatic for long periods but can eventually cause severe pain, nausea, and jaundice. Affecting over 700 million people globally, the disease is increasingly common in urbanized regions due to lifestyle changes, including poor diet and rising obesity. Its high prevalence makes it a significant contributor to healthcare costs, especially as a leading cause of emergency surgeries, highlighting the importance of early risk detection.

Multiple risk factors contribute to gallstone formation, including age, female sex, obesity, rapid weight loss, diabetes, and genetic predisposition. Body composition indicators such as visceral fat, total body fat, and muscle mass are also critical, as they influence metabolic processes and bile composition. Advances in bioelectrical impedance analysis allow for quick, noninvasive measurement of these parameters. When combined with biochemical markers from blood tests, these data provide a comprehensive feature set that can be leveraged for machine learning-based risk prediction.

The dataset used in this study consists of clinical and bioimpedance data from 319 individuals, including 161 diagnosed cases of gallstones, with 38 features and a binary target variable. This structured dataset supports exploratory data analysis, statistical testing, and predictive modeling. Machine learning methods such as logistic regression and random forests are particularly suitable, offering both interpretability and accuracy. The study applies these techniques to generate reproducible insights and develop practical tools for early screening and risk assessment of gallstone disease.

Problem Statement

Current clinical methods for identifying gallstone risk rely on patient history, physical exams, imaging, and basic blood tests, which are effective for symptomatic cases but limited for early detection in seemingly healthy individuals. As a result, many patients are diagnosed only after acute complications arise. Traditional approaches also focus on a narrow set of risk factors, often overlooking broader metabolic and body composition indicators.

This study addresses these limitations by using exploratory analysis, hypothesis testing, and machine learning on a clinical dataset to identify key differences between patients with and without gallstones. It evaluates body composition trends, tests associations with obesity, and develops predictive models. The outcome is a set of reproducible insights aimed at supporting low cost, early screening tools for high-risk populations.

Purpose of the Study

The purpose of this study is to analyze the Gallstone dataset using exploratory data analysis, formal statistical testing, and simple machine learning models that can run on commodity hardware. The work aims to achieve four objectives. First, it explores how clinical, biochemical, and bioimpedance features are distributed and how their values differ between patients with gallstones and patients without gallstones. Second, it studies how body composition metrics vary across body mass index categories from underweight to obese, looking for monotonic trends that may reveal a metabolic gradient. Third, it tests whether obesity, defined as a body mass index of thirty kilograms per square meter or higher, is statistically associated with the presence of gallstones. Fourth, it builds two

interpretable classification models, namely logistic regression and random forest, to predict gallstone status and to identify the most important features that drive this prediction.

Scope and objectives

The scope of this study is limited to the Gallstone dataset (UCI Machine Learning Repository, Dataset ID 1150), which contains 319 patient records and thirty-eight clinical, biochemical, and bioimpedance features along with a binary target indicating gallstone presence. The analysis focuses on supervised binary classification and descriptive statistical exploration using Python based open-source libraries such as pandas, numpy, scipy, matplotlib, seaborn, and scikit learn. The study does not seek to develop a clinical tool for direct patient use. Rather it produces validated proof of concept models and evidence-based observations that may inform the future design of clinical decision support systems and low-cost screening protocols. The four research questions that guide the study are stated below. The null and alternative hypotheses for each research question are described in detail in the section titled **“Solution to RQ1 to RQ4”**.

Research Question 1 (RQ1): What patterns exist in the distribution of clinical, biochemical, and bioimpedance features of the Gallstone dataset, and how do key variables differ between patients with gallstones and patients without gallstones?

Research Question 2 (RQ2): How do body composition metrics such as total body fat ratio, visceral fat rating, lean mass percentage, and muscle mass vary across body mass index categories ranging from underweight to obese, and what observable trends emerge as body mass index increases?

Research Question 3 (RQ3): Is there a statistically significant association between obesity status, defined as a body mass index of thirty kilograms per square meter or higher, and the presence of gallstones in this patient population?

Research Question 4 (RQ4): Can simple machine learning classification models accurately predict whether a patient has gallstones based on their clinical, biochemical, and bioimpedance features, and which features are the most important drivers of this prediction?

Sample size calculation

To ensure that the study is powered adequately to detect meaningful effects and to validate model performance with statistical confidence, minimum sample sizes were calculated for each research question using established methods from statistical power analysis and confidence interval estimation. All calculations assume a two-tailed significance level of $\alpha = 0.05$ and, where applicable, a desired statistical power of $1 - \beta = 0.80$.

RQ1: Confidence Interval for a proportion: A confidence interval approach was used to estimate the proportion of patients with gallstones in the dataset within a margin of error of 0.055. Using a conservative estimated proportion of 0.50, which gives the largest required sample, the standard sample size formula for a proportion is shown below:

$$n = \frac{Z^2 \cdot p \cdot (1 - p)}{e^2}$$

After substituting the values into the formula:

$$n = \frac{(1.96)^2 \times 0.50 \times (1 - 0.50)}{(0.055)^2} = 317.32 \approx 318$$

RQ2: Confidence Interval for a mean: A confidence interval approach was used to estimate the mean body mass index value within each body composition category with a margin of error of 0.60 kilograms per square meter and an estimated standard deviation of 5.00 kilograms per square meter. The formula is given below:

$$n = \left(\frac{Z \cdot \sigma}{e} \right)^2$$

where Z equals 1.96 for ninety five percent confidence, sigma equals 5.00 kilograms per square meter for an estimated standard deviation of body mass index drawn from clinical knowledge of this patient population, and e equals 0.60 kilograms per square meter for the desired margin of error.

After substituting the values into the formula:

$$n = \left(\frac{1.96 \times 5.0}{0.60} \right)^2 = 266.78 \approx 267$$

RQ3: Chi-square power analysis: This involves a formal hypothesis test comparing gallstone rates between two groups defined by obesity status. The goal is to enroll enough patients so the test can detect a real difference if one exists. The formula is given below.

$$n = \left(\frac{z_{\alpha/2} + z_{\beta}}{\omega} \right)^2$$

where $z_{\alpha/2} = 1.96$ (critical value at $\alpha = 0.05$, two tailed), $z_{\beta} = 0.84$ (critical value for 80% power), and $\omega = 0.30$ (Cohen's w, a medium effect size for a chi-square test).

After substituting the values into the formula:

$$n = \left(\frac{1.96 + 0.84}{0.30} \right)^2 = 87.07$$

Hence, 87 patients per group is required, which gives 174 total samples across both obesity groups.

RQ4: Logistic regression power analysis: A logistic regression power analysis can be used. The minimum sample size needs to account for both the desired effect size and the number of predictors. The formula used comes from Cohen framework for multiple regression:

$$n = \left(\frac{L}{f^2} \right) + u + 1$$

where $L = 7.85$ is a tabled non-centrality parameter for power = 0.80 at $\alpha = 0.05$, $f^2 = 0.15$ is a medium effect size for regression, and $u = 38$ is the number of predictor variables in the dataset.

After substituting the values into the formula:

$$n = \left(\frac{7.85}{0.15} \right) + 38 + 1 \approx 92$$

A minimum of 92 observations is required to build a reliable logistic regression model with 38 predictors, ensuring sufficient power to detect a medium effect size.

Table 1

Summary of Sample Size Calculations

Research Question	Method Used	Key Parameters	Minimum Sample Size (N)
RQ1	Confidence Interval (Estimating proportion of gallstone cases)	$\alpha = 0.05$, Confidence level = 95 percent, $Z = 1.96$, percent so $Z = 1.96$, Estimated proportion $p = 0.50$, Margin of error $e = 0.055$	$N = 318$
RQ2	Confidence Interval (Estimating mean body mass index)	$\alpha = 0.05$, Confidence level = 95 percent, $Z = 1.96$, Standard deviation $\sigma = 5.00$ kilograms per square meter, Margin of error $e = 0.60$ kilograms per square meter	$N = 267$
RQ3	Power Analysis (Chi-square test for independence)	$\alpha = 0.05$, $z_{\alpha/2} = 1.96$, Power 80 percent i.e. $z_{\beta} = 0.84$, Effect size $\omega = 0.30$	$N = 174$
RQ4	Power Analysis (Logistic Regression, Cohen f^2)	$L = 7.85$ for $\alpha = 0.05$, Power = 0.80, $f^2 = 0.15$ medium effect size, $u = 38$ predictor variables	$N = 92$

The final sample size for this study is chosen as the maximum of the minimum values computed across all four research questions, which is 318. The Gallstone dataset contains 319 total patient records with mostly complete feature data. Since 319 meets the required minimum of 318, the dataset is sufficient to support all planned analyses at the targeted precision and power levels.

Data Description

The dataset used in this study is the Gallstone dataset published on the University of California Irvine Machine Learning Repository. It is publicly available at the following address: <https://archive.ics.uci.edu/dataset/1150/gallstone-1>. The dataset is provided as a single comma separated values file and can be loaded directly into Python using the pandas library (McKinney, 2010). It is also fully compatible with the scikit learn machine learning library (Pedregosa et al., 2011), which will be used to build and evaluate all classification models in this study. The dataset was collected at Ankara VM Medical Park Hospital. Each patient record carries a binary label that indicates whether gallstones were detected during the imaging examination.

Data dictionary

Data Dictionary for the Gallstone Disease dataset:

Feature	Description	Data Type	Values / Units
Gallstone Status	Outcome label indicating whether gallstones were detected during ultrasound	Binary	1 = Gallstone present, 0 = No gallstone
Age	Patient age at the time of evaluation	Integer	Years
Gender	Biological sex of the patient	Categorical	0 = Male, 1 = Female
Comorbidity	Number of associated chronic conditions present	Ordinal	0 to 3
Coronary Artery Disease	Indicates whether the patient has been diagnosed with coronary artery disease	Binary	1 = Yes, 0 = No

Feature	Description	Data Type	Values / Units
Hypothyroidism	Indicates whether the patient has hypothyroidism	Binary	1 = Yes, 0 = No
Hyperlipidemia	Indicates whether the patient has elevated blood lipid levels	Binary	1 = Yes, 0 = No
Diabetes Mellitus	Indicates whether the patient has diabetes mellitus	Binary	1 = Yes, 0 = No
Height	Patient height	Continuous	Centimeters
Weight	Patient weight	Continuous	Kilograms
Body Mass Index	Calculated body mass index value	Continuous	Kilograms per square meter
Total Body Water	Total body water content from bioimpedance analysis	Continuous	Liters
Extracellular Water	Water located outside the cells from bioimpedance analysis	Continuous	Liters
Intracellular Water	Water located inside the cells from bioimpedance analysis	Continuous	Liters
Extracellular Fluid Ratio	Ratio of extracellular fluid to total body water	Continuous	Decimal value
Total Body Fat Ratio	Percentage of body weight that is fat tissue	Continuous	Percent
Lean Mass	Percentage of body weight that is lean tissue	Continuous	Percent
Body Protein Content	Percentage of body weight that is protein tissue	Continuous	Percent
Visceral Fat Rating	Index that estimates fat surrounding internal organs	Ordinal	1 to 30 scale

Feature	Description	Data Type	Values / Units
Bone Mass	Estimated mass of bone tissue	Continuous	Kilograms
Muscle Mass	Estimated mass of skeletal muscle tissue	Continuous	Kilograms
Obesity Percent-age	Computed obesity index relative to ideal body weight	Continuous	Percent
Total Fat Content	Total mass of body fat tissue	Continuous	Kilograms
Visceral Fat Area	Estimated area of visceral fat in cross sectional view	Continuous	Square centimeters
Visceral Muscle Area	Estimated area of visceral muscle tissue	Continuous	Kilograms
Hepatic Fat Accumulation	Estimated fat accumulation in the liver	Ordinal	0 to 4 grade
Glucose	Fasting blood glucose level	Continuous	Milligrams per deciliter
Total Cholesterol	Serum total cholesterol level	Continuous	Milligrams per deciliter
Low Density Lipoprotein	Serum low density lipoprotein cholesterol level	Continuous	Milligrams per deciliter
High Density Lipoprotein	Serum high density lipoprotein cholesterol level	Continuous	Milligrams per deciliter
Triglyceride	Serum triglyceride level	Continuous	Milligrams per deciliter
Aspartate Aminotransferase	Liver enzyme that indicates hepatic cell injury	Continuous	International units per liter
Alanine Aminotransferase	Liver enzyme is commonly used to assess liver function	Continuous	International units per liter
Alkaline Phosphatase	Enzyme associated with bile duct activity	Continuous	International units per liter

Feature	Description	Data Type	Values / Units
Creatinine	Serum creatinine level used to assess kidney function	Continuous	Milligrams per deciliter
Glomerular Filtration Rate	Estimated rate of fluid filtration through the kidneys	Continuous	Milliliters per minute per 1.73 square meters
C Reactive Protein	Acute phase protein indicating systemic inflammation	Continuous	Milligrams per liter
Hemoglobin	Concentration of hemoglobin in the blood	Continuous	Grams per deciliter
Vitamin D	Serum 25 hydroxy vitamin D level	Continuous	Nanograms per milliliter

The dataset includes 319 patient records with 39 variables in total, covering demographic, comorbidity, body composition, biochemical, and outcome information. The primary outcome variable is Gallstone Status, where 1 indicates a confirmed gallstone case and 0 indicates a control patient without gallstones. Missing biochemical and bioimpedance values are addressed using median imputation for continuous variables and mode imputation for categorical variables before any modeling step.

Analytic approach

The analytic approach for this study is organized according to the type of research question. RQ1 combines exploratory data analysis with formal hypothesis testing, applying the Mann–Whitney U test to each continuous feature and the chi-square test of independence to each categorical feature to compare patients with gallstones against patients without gallstones. RQ2 relies on exploratory data analysis to identify patterns and relationships using visual and descriptive methods, without formal

statistical inference. RQ3 employs hypothesis testing with clearly defined null and alternative hypotheses and a decision rule. RQ4 uses simple machine learning classification models.

For the modeling component, the dataset is split into training and testing sets using an 80 to 20 ratio with stratified sampling to preserve class balance. Preprocessing includes feature standardization using zero mean and unit variance scaling. Model performance is assessed using five fold cross validation on the training data to ensure reliable estimates, while the test set is reserved for a single final evaluation.

Solution to RQ1-4

RQ1: Feature Distribution by Gallstone Status

This research question can be addressed through exploratory data analysis supported by formal statistical hypothesis testing. The goal is to understand how clinical, biochemical, and bioimpedance features are distributed in the dataset and to identify statistically significant differences between patients with gallstones and patients without gallstones.

Null Hypothesis (H_0)

For every continuous clinical, biochemical, and bioimpedance feature (including body mass index, total body fat ratio, visceral fat rating, glucose, low density lipoprotein, high density lipoprotein, alanine aminotransferase, and C reactive protein), the distribution is the same in patients with gallstones and patients without gallstones. For every categorical feature (including gender, comorbidity

count, coronary artery disease, hypothyroidism, hyperlipidemia, and diabetes mellitus), gallstone status is independent of that feature.

Alternative Hypothesis (H_1)

At least one continuous feature shows a statistically significant difference in its distribution between patients with gallstones and patients without gallstones (Mann–Whitney U test, $p < 0.05$), or at least one categorical feature shows a statistically significant association with gallstone status (chi-square test of independence, $p < 0.05$).

Approach

The exploratory data analysis examines patterns in the gallstone dataset through a structured set of visualizations and summaries. A bar chart shows the distribution of patients across gallstone status categories. Box plots and violin plots compare key continuous variables between patients with and without gallstones. Grouped bar charts illustrate outcome proportions across categorical variables including gender, comorbidity count, and chronic disease indicators. A Spearman correlation heatmap identifies relationships among continuous features, while a missing value summary highlights data gaps prior to imputation. Insights from these analyses are synthesized into a narrative to guide further investigation.

Following EDA, statistical hypothesis testing is conducted at a significance level of 0.05 to evaluate the research question. The Mann–Whitney U test is used for continuous variables to compare distributions between the two patient groups. For categorical variables, the chi-square test of

independence assesses associations with gallstone status, with Fisher's exact test applied when expected cell counts are below five.

Expected Output

Exploratory data analysis will produce visual summaries including distribution plots, box and violin plots, bar charts, a correlation heatmap, and a missing value table. Two inferential summary tables will also be created. The first will report Mann–Whitney U statistics, raw and adjusted p values, and effect sizes for continuous features. The second will report chi-square or Fisher test results, degrees of freedom, p values, and association measures for categorical features, with significance flagged at adjusted $p < 0.05$. Results are expected to show higher BMI, body fat, triglycerides, visceral fat, and glucose in gallstone patients, along with significant categorical associations.

RQ2: Body Composition Trends Across Body Mass Index Categories

This research question can be addressed entirely through exploratory data analysis. The goal is to observe how key body composition metrics change as patients move from the underweight category through the obese category, and to highlight any visible monotonic trend or threshold effect that may be clinically meaningful.

Null Hypothesis (H_0)

No observable monotonic trend is expected between body mass index category and body composition metric values, and the median values of the body composition metrics are expected to look

similar across the four body mass index categories of underweight, normal, overweight, and obese when viewed through descriptive plots and summary statistics.

Alternative Hypothesis (H_1)

At least one body composition metric shows an observable monotonic trend, either positive or negative Spearman rank correlation, with the body mass index category, or its median value differs visibly across the four categories as revealed by exploratory box plots, violin plots, and median heatmaps.

Approach

The analysis explores how body composition metrics vary across body mass index categories defined by World Health Organization cutoffs: underweight, normal, overweight, and obese. A bar chart presents the distribution of patients across these groups. Box and violin plots illustrate changes in key continuous metrics, including body fat ratio, lean mass percentage, body protein, visceral fat rating, muscle mass, total fat content, and visceral fat area across categories. A heatmap of median values provides a consolidated view of trends across all metrics. Additionally, Spearman rank correlations between each metric and the ordered BMI categories quantify the direction and strength of relationships, showing whether metrics increase or decrease with BMI.

Expected Output

The exploratory data analysis will generate category wise visualizations including a category distribution chart, box and violin plots for body composition metrics, a median value heatmap, and

Spearman rank correlation summaries. It is expected to show increasing trends in total body fat ratio, visceral fat rating, total fat content, and visceral fat area, alongside decreasing or stable trends in lean mass percentage and body protein content as body mass index moves from the underweight category to the obese category. These observations will be supported by descriptive narrative insights without inferential statistics.

RQ3: Association Between Obesity and Gallstone Presence

This research question can be addressed through formal statistical hypothesis testing. The hypothesis to be tested is stated below.

Null Hypothesis (H_0):

The proportion of patients with gallstones in the obese group is equal to the proportion of patients with gallstones in the non-obese group ($p_1 = p_2$).

Alternative Hypothesis (H_1):

The proportion of patients with gallstones in the obese group is not equal to the proportion of patients with gallstones in the non-obese group ($p_1 \neq p_2$).

Approach

The test will be conducted at a significance level of alpha equal to 0.05 using a two tailed chi square test for independence. If the p value is less than 0.05, the null hypothesis will be rejected.

Each patient will first be classified as obese if their body mass index is thirty kilograms per square meter or higher, and as non-obese otherwise. Before conducting the test, the exploratory data analysis stage for RQ3 will include a grouped bar chart showing the number and proportion of patients with and without gallstones within each obesity group. This visual check will confirm whether any obvious imbalance exists between the two groups before the formal test is applied.

The test will be applied to a 2x2 contingency table with obesity status as the rows and gallstone status as the columns. The chi square statistics are computed as shown below.

$$\text{Sum of } \frac{(\text{Observed} - \text{Expected})^2}{\text{Expected}} \text{ (over all four cells)}$$

The odds ratio and its ninety five percent confidence interval will be calculated to measure the direction and strength of the association. Fisher exact test will be used as a backup if any cell count in the contingency table falls below five. The decision and interpretation of the test will be reported clearly in the final report along with the practical clinical meaning of the effect size.

Expected Output

This research question will generate inferential results such as a 2x2 contingency table, expected counts, p value, and odds ratio with a 95 percent confidence interval. A chi square test will be used to assess independence, reporting statistics, degrees of freedom, and p value. If expected counts are below five, Fisher exact test will replace it. Based on prior evidence, the p value is expected to be below 0.05, indicating a statistically significant association between obesity and gallstone presence.

RQ4: Gallstone Status Prediction Using Machine Learning

Null Hypothesis (H_0):

The trained classification model performs no better than chance in predicting gallstone status, that is the population area under the receiver operating characteristic curve is less than or equal to 0.50, and equivalently the model mean cross validated area under the curve is not statistically distinguishable from 0.50.

Alternative Hypothesis (H_1):

The trained classification model performs better than chance, that is the population area under the receiver operating characteristic curve is greater than 0.50.

Approach

This research question is addressed using binary classification with logistic regression and random forest models. The target variable is Gallstone Status, where 1 indicates presence and 0 indicates absence. Missing values are handled through median imputation for continuous features and mode imputation for categorical features, while continuous variables are standardized. If class imbalance exists, synthetic minority oversampling is applied only to the training data. The dataset is split into 80 percent training and 20 percent testing sets using stratified sampling, with five fold cross validation on the training data. Logistic regression provides an interpretable baseline, while random forest captures nonlinear patterns and feature importance. Both models are evaluated using accuracy, precision, recall, F1 score, and AUC, with feature importance highlighting key risk factors.

Evaluation Metrics:

Accuracy:

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

Precision:

$$\text{Precision} = TP / (TP + FP)$$

Recall:

$$\text{Recall} = TP / (TP + FN)$$

F1 Score:

$$F1 = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

Expected Output

This research will generate a cleaned dataset, a stratified train–test split, and two trained models. Key outputs include cross validated AUC scores, confusion matrices, ROC curves, classification reports, and a ranked list of important features. The random forest model is anticipated to outperform logistic regression in predictive performance, while logistic regression remains valuable as an interpretable baseline for understanding feature effects.

Recommendation and application

This research primarily benefits gastroenterologists, hepatologists, internal medicine physicians, and primary care providers by helping them identify key clinical, biochemical, and body composition factors associated with gallstone disease. The findings support more informed decisions about which patients should undergo diagnostic imaging and enhance understanding of how metabolic risk changes with increasing body weight, aiding preventive care and lifestyle counseling.

A simple prediction model based on routine measurements can serve as an effective screening tool in outpatient settings, especially where imaging access is limited. It enables early identification of high-risk patients, potentially reducing emergency cases. Additionally, clinical data scientists can use the study's reproducible notebooks to develop decision support systems and adapt models to local populations.

The study is also valuable for public health researchers and policymakers by emphasizing the link between obesity and gallstone risk. It recommends standardized recording of key health indicators to improve future predictive models.

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